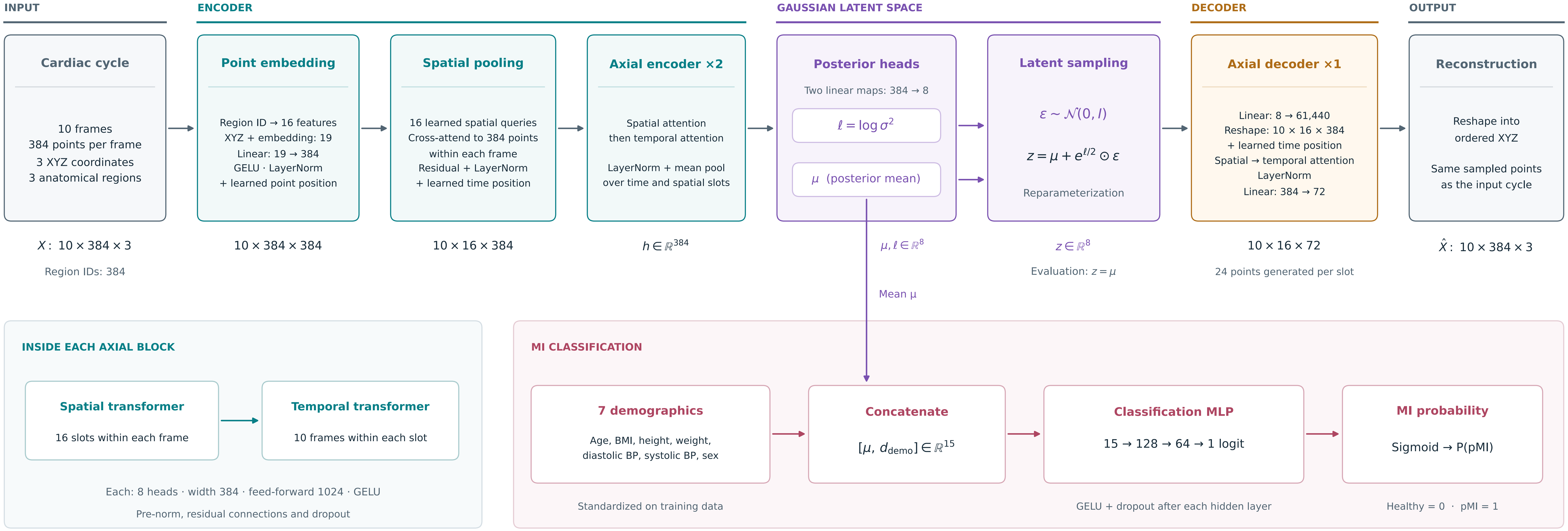
Cardiac variational autoencoder

MODEL ARCHITECTURE

Spatial–temporal attention encodes a cardiac cycle; the latent mean also supports MI prediction.



INPUT

Cardiac cycle

10 frames
384 points per frame
3 XYZ coordinates
3 anatomical regions

$X:\ 10 \times 384 \times 3$

Region IDs: 384

ENCODER

Point embedding

Region ID → 16 features
XYZ + embedding: 19
Linear: 19 → 384
GELU · LayerNorm
+ learned point position

$10 \times 384 \times 384$

Spatial pooling

16 learned spatial queries
Cross-attend to 384 points
within each frame
Residual + LayerNorm
+ learned time position

$10 \times 16 \times 384$

Axial encoder ×2

Spatial attention
then temporal attention

LayerNorm + mean pool
over time and spatial slots

$h \in \mathbb{R}^{384}$

GAUSSIAN LATENT SPACE

Posterior heads

Two linear maps: 384 → 8

$\ell = \log \sigma^2$

μ (posterior mean)

$\mu, \ell \in \mathbb{R}^8$

Mean μ

Latent sampling

$\varepsilon \sim \mathcal{N}(0, I)$

$z = \mu + e^{\ell/2} \odot \varepsilon$

Reparameterization

$z \in \mathbb{R}^8$

Evaluation: $z = \mu$

DECODER

Axial decoder ×1

Linear: 8 → 61,440
Reshape: 10 × 16 × 384
+ learned time position
Spatial → temporal attention
LayerNorm
Linear: 384 → 72

$10 \times 16 \times 72$

24 points generated per slot

OUTPUT

Reconstruction

Reshape into
ordered XYZ

Same sampled points
as the input cycle

$\hat{X}:\ 10 \times 384 \times 3$

INSIDE EACH AXIAL BLOCK

Spatial transformer

16 slots within each frame

Temporal transformer

10 frames within each slot

Each: 8 heads · width 384 · feed-forward 1024 · GELU

Pre-norm, residual connections and dropout

MI CLASSIFICATION

7 demographics

Age, BMI, height, weight,
diastolic BP, systolic BP, sex

Standardized on training data

Concatenate

$[\mu,\ d_{\text{demo}}] \in \mathbb{R}^{15}$

Classification MLP

15 → 128 → 64 → 1 logit

GELU + dropout after each hidden layer

MI probability

Sigmoid → P(pMI)

Healthy = 0 · pMI = 1

TRAINING OBJECTIVE

$$\mathcal{L} = \mathrm{MSE}(X, \hat{X}) + \beta\, \overline{D_{\mathrm{KL}}}(q_\phi(z \mid X, r) \,\|\, \mathcal{N}(0, I)) + \lambda\, \mathrm{BCEWithLogits}(y, a)$$

Reconstruction + KL
+ supervised classification

MSE averages all XYZ values; KL averages patients and latent dimensions. Default training: β warms to 0.01 over 10 epochs; λ = 1; BCE uses training-derived class weighting.

Saved model: latent dimension 8, dropout 0.1. Tensor shapes omit the batch dimension. Demographics enter the classification head only.

Source: src/mcpp/vae.py